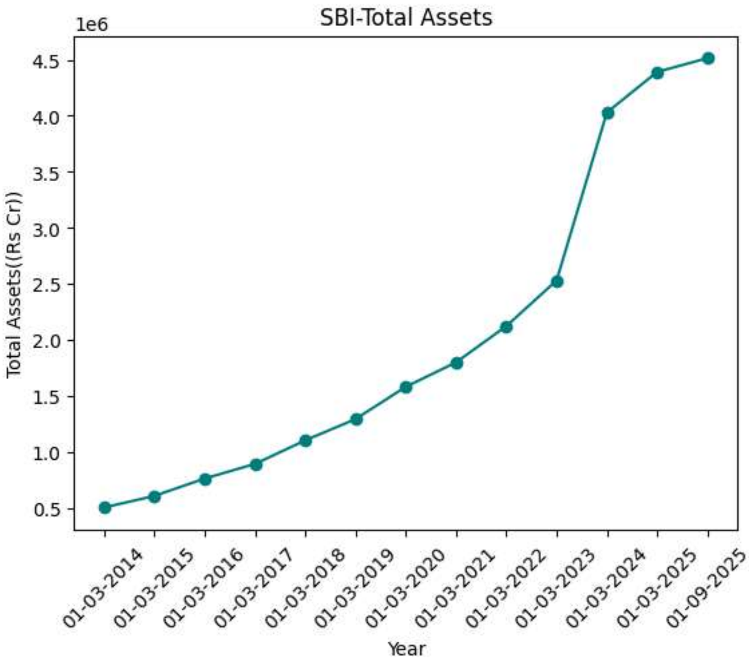
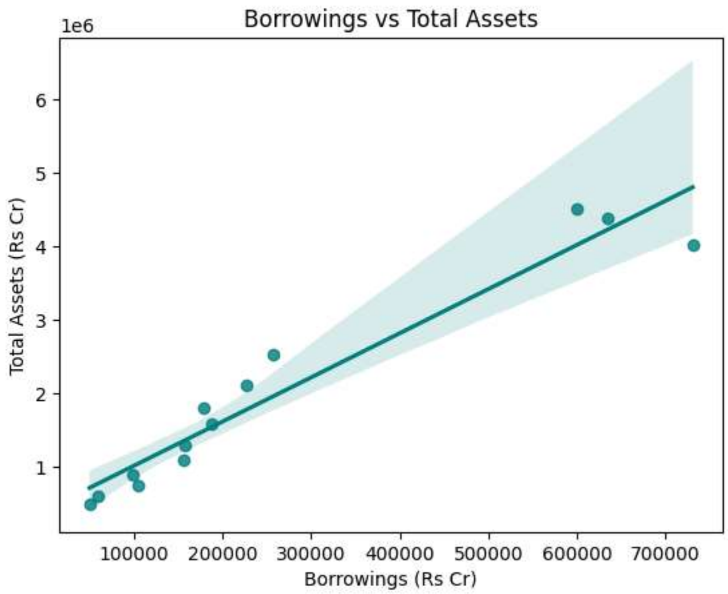
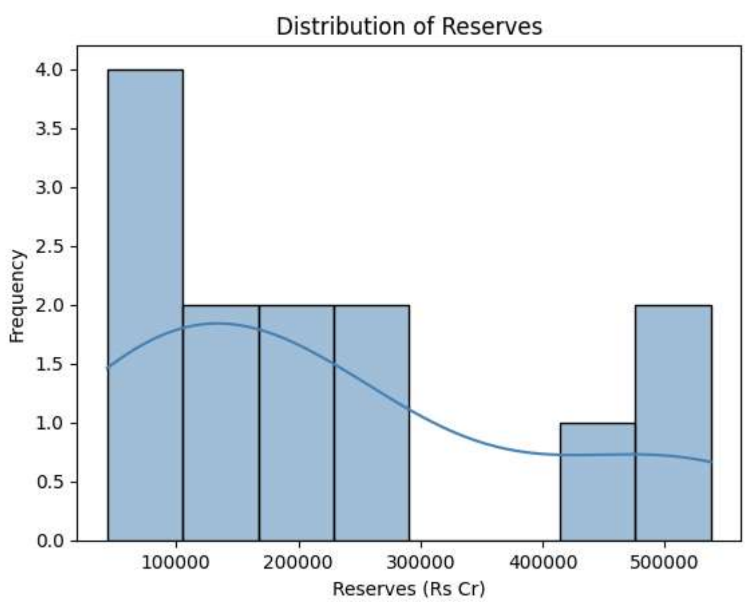
```
#Aanya Aggarwal, 12520984
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv("/content/sbibs.csv")
df.sort_values('Year',inplace=True)
plt.plot(df['Year'], df['TotalAssets'], marker='o', color='teal')
plt.title('SBI-Total Assets')
plt.xlabel('Year')
plt.ylabel('Total Assets((Rs Cr))')
plt.xticks(rotation=45)
plt.show()
```




```
#Aanya Aggarwal, 12520984
from pathlib import Path
#Aanya Aggarwal, 12520984
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
Path="/content/sbibs.csv"
df=pd.read_csv(Path, parse_dates=['Year'])
sns.regplot(x='Borrowing', y='TotalAssets', data=df, color='teal')
plt.title('Borrowings vs Total Assets')
plt.xlabel('Borrowings (Rs Cr)')
plt.ylabel('Total Assets (Rs Cr)')
plt.show()
```

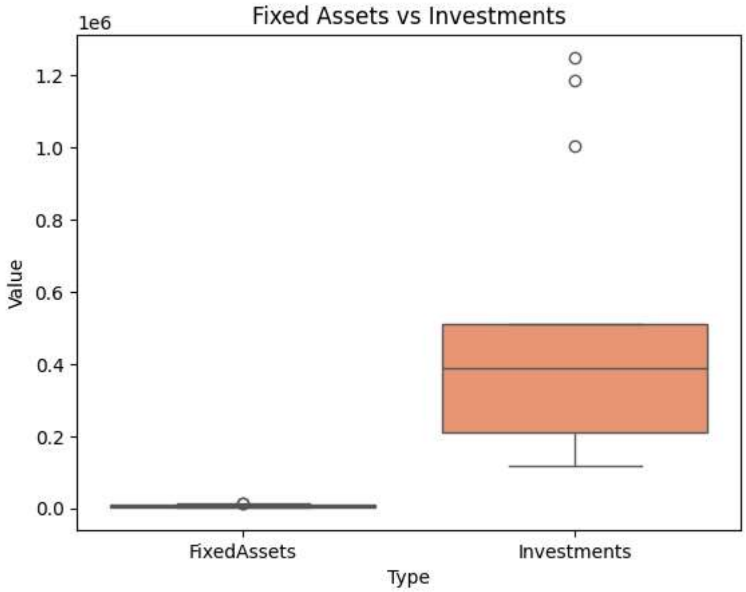



```
#Aanya Aggarwal, 12520984
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
Path="/content/sbibs.csv"
df=pd.read_csv(Path, parse_dates=['Year'])
sns.histplot(df['Reserves'], kde=True, color='steelblue', bins=8)
plt.title('Distribution of Reserves')
plt.xlabel('Reserves (Rs Cr)')
plt.ylabel('Frequency')
plt.show()
```



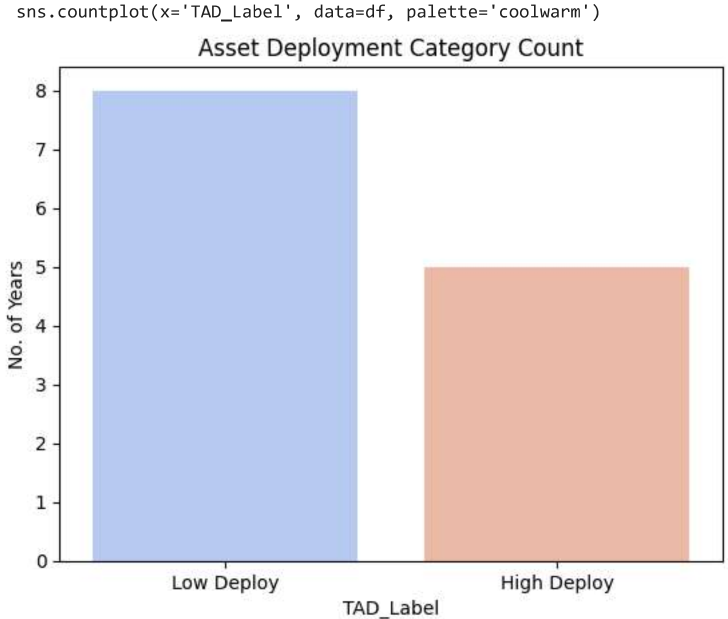

```
#Aanya Aggarwal, 12520984
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
Path="/content/sbibs.csv"
df=pd.read_csv(Path, parse_dates=['Year'])
cols=['FixedAssets', 'Investments']
df_m=df[cols].melt(var_name='Type', value_name="Value")
sns.boxplot(x='Type', y="Value", data=df_m, palette='Set2')
plt.title('Fixed Assets vs Investments')
plt.show()
```

```
/tmp/ipykernel_7439/716291215.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.boxplot(x='Type', y="Value", data=df_m, palette='Set2')
```



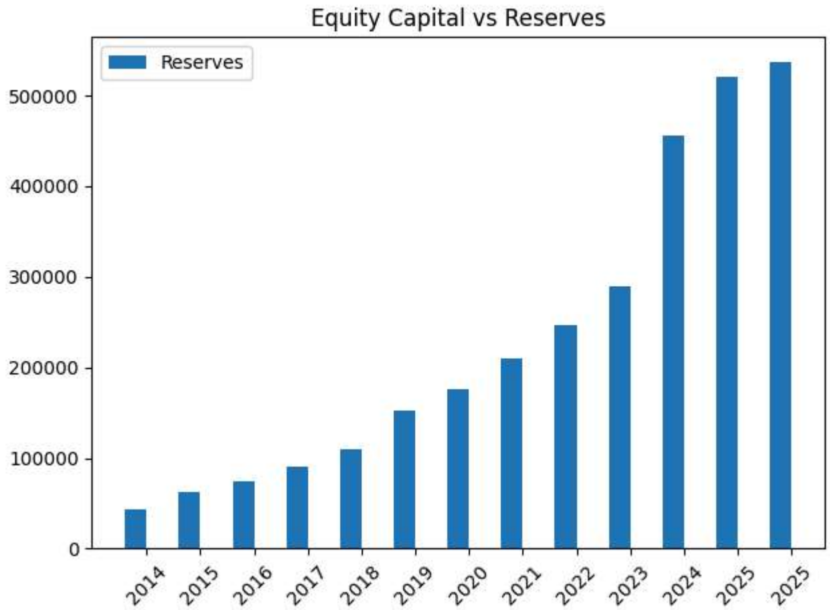

```
#Aanya Aggarwal, 12520984
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
Path="/content/sbibs.csv"
df=pd.read_csv(Path, parse_dates=['Year'])
df['TAD_Label']=df['TAD'].map({0:'Low Deploy', 1:'High Deploy'})
sns.countplot(x='TAD_Label', data=df, palette='coolwarm')
plt.title('Asset Deployment Category Count')
plt.ylabel('No. of Years')
plt.show()
```

```
/tmp/ipykernel_7439/3504127184.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.countplot(x='TAD_Label', data=df, palette='coolwarm')
```

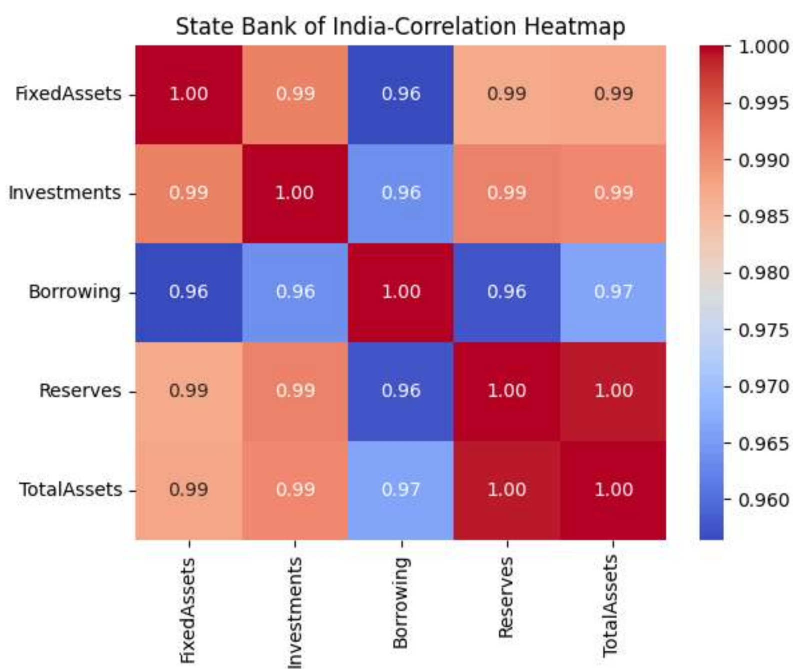



```
#Aanya Aggarwal, 12520984
import pandas as pd
```

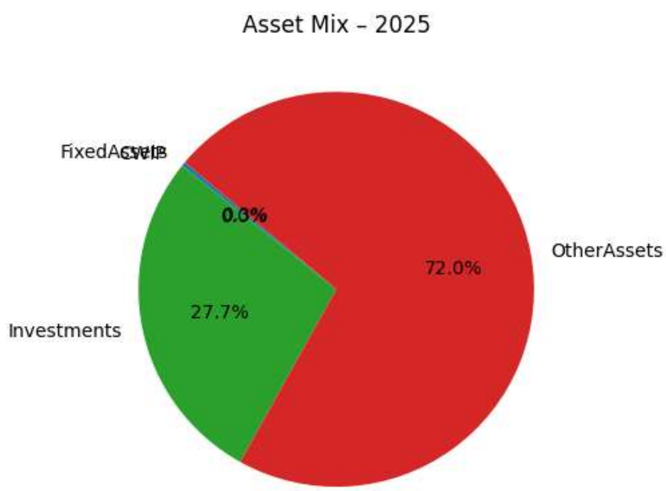
```
import matplotlib.pyplot as plt
import numpy as np
Path="/content/sbibs.csv"
df=pd.read_csv(Path, parse_dates=['Year'])
x=np.arange(len(df))
plt.bar(x-0.2, df['Reserves'], width=0.4, label="Reserves")
plt.xticks(x, df['Year'].dt.year, rotation=45)
plt.title('Equity Capital vs Reserves')
plt.legend(); plt.tight_layout()
plt.show()
```



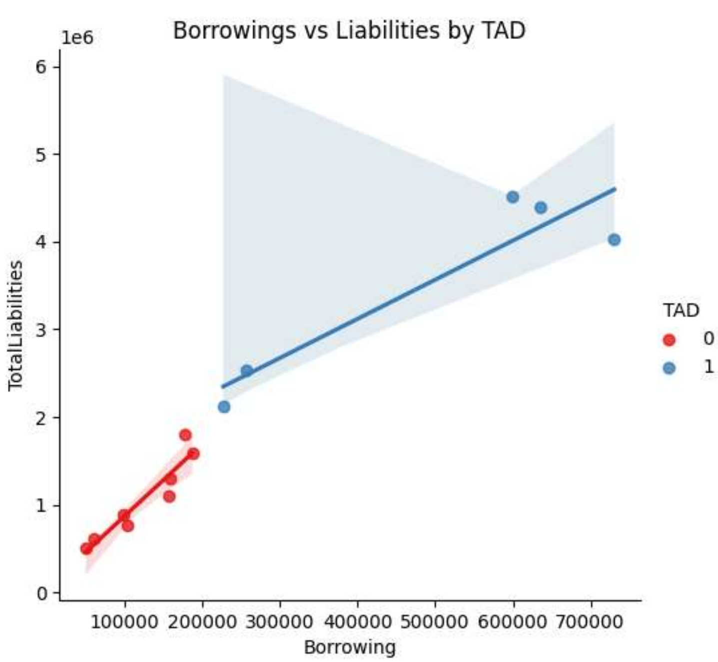

```
#Aanya Aggarwal, 12520984
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
PATH= r'/content/sbibs.csv'
df=pd.read_csv(PATH, parse_dates=['Year'])
cols=['FixedAssets', 'Investments','Borrowing', 'Reserves', 'TotalAssets']
sns.heatmap(df[cols].corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('State Bank of India-Correlation Heatmap')
plt.show()
```



```
#Aanya Aggarwal, 12520984
import pandas as pd
import matplotlib.pyplot as plt
PATH = r'/content/sbibs.csv'
df   = pd.read_csv(PATH, parse_dates=['Year'])
latest = df.iloc[-1]
labels = ['FixedAssets','CWIP', 'Investments','OtherAssets']
values = [latest[l] for l in labels]
plt.pie(values, labels=labels, autopct='%1.1f%%', startangle=140)
plt.title(f'Asset Mix – {latest["Year"].year}')
plt.show()
```




```
#Aanya Aggarwal, 12520984
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
PATH = r'/content/sbibs.csv'
df   = pd.read_csv(PATH, parse_dates=['Year'])
df['TAD'] = df['TAD'].astype(str)
sns.lmplot(x='Borrowing', y='TotalLiabilities', data=df, hue='TAD', palette='Set1')
plt.title('Borrowings vs Liabilities by TAD')
plt.show()
```




```
#Aanya Aggarwal, 12520984
import pandas as pd
import numpy as np
```

```
PATH = r'/content/sbibs.csv'
df   = pd.read_csv(PATH, parse_dates=['Year'])
x = df['Borrowing'].values
y = df['TotalAssets'].values
r = np.corrcoef(x, y)[0, 1]
if r > 0.7:
    print('Debt-driven growth – REVIEW leverage')
elif r > 0.4:
    print('Moderate link – MONITOR borrowings')
else:
    print('Healthy – assets not debt-dependent')
print(f'Correlation = {r:.3f}')
```